DATASCI 285 — Final Project Topics

There are six topics to choose from. You'll work in a group of two or three. Each topic centers on one study, and most topics also have one or more papers that complicate it. For five of the six topics, we've checked the data and we can give it to you. For the sixth, the data isn't available. You work from the authors' published tables, code and output, and from simulations you build yourself. Working without the data is what that topic is about.

You aren't expected to find errors. If you establish that a study's methods were sound, that's as good an outcome as finding a problem with them.

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1. Does anti-LGBTQ political messaging win elections?

Papers.

- Violeta I. Haas, Konstantin Bogatyrev, Tarik Abou-Chadi, Heike Klüver and Lukas F. Stoetzer. The Electoral Effects of State-Sponsored Anti-LGBTQ Measures. *The Journal of Politics*, forthcoming. Free copy. Data and code: Harvard Dataverse.
- Phillip M. Ayoub, Douglas Page and Samuel Whitt. Anti-LGBTIQ Rhetoric and Electoral Outcomes under the Shadow of War: Evidence from Poland's 2023 Parliamentary Election. *European Journal of Political Research* 64(4), 2025. Data and code: Harvard Dataverse.

Summary. Between 2019 and 2021, 109 Polish local and regional governments passed resolutions against "LGBT ideology." Haas and colleagues ask whether these resolutions changed election results, using municipal returns from six parliamentary elections. Ayoub and colleagues ask a related question with a randomized survey experiment. They surveyed 1,000 Polish voters during the war in Ukraine. You get both datasets and the authors' code.

Challenges. The survey experiment is a good place to start. It has four randomly assigned groups and a yes-or-no outcome, so the analysis is a clean difference in proportions. The observational study is harder. Resolutions were passed at three different levels of government,

so when a county passed one, it treated every municipality inside that county at once. That means there are hundreds of municipalities but far fewer independent decisions. What does that do to the uncertainty? The comparison group is a question too. In 87 councils the same resolution was debated and voted down, and those councils may be a better comparison group than untreated municipalities in general.

2. What do résumé experiments tell us about discrimination?

Papers.

- Marianne Bertrand and Sendhil Mullainathan. Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination. *American Economic Review* 94(4), 2004. Free working-paper version. Data: openICPSR.
- Issa Kohler-Hausmann. Eddie Murphy and the Dangers of Counterfactual Causal Thinking About Detecting Racial Discrimination. *Northwestern University Law Review* 113(5), 2019.
- Alexander Williams Tolbert (Data and Decision Sciences, Emory). Causal Agnosticism About Race: Variable Selection Problems in Causal Inference. *Philosophy of Science* 91(5), 2024. Open access.

Summary. Bertrand and Mullainathan sent nearly 5,000 fictitious résumés to real employers. Each résumé was randomly given either a name that sounds white or a name that sounds Black, and the researchers recorded which résumés got callbacks. Because the name was randomly assigned, the comparison between the two groups of names is clean, and the gap in callbacks is large. Kohler-Hausmann asks what quantity that experiment estimates. To talk about a "treatment effect of race," you have to treat race as something you can vary while holding the person otherwise fixed. In practice, that means reducing race to a name or some other visible marker. Her argument is that this isn't a practical limitation to work around. It's a problem with the estimand itself.

Challenges. Callbacks are uncommon, and that drives how large a study like this has to be, so a power calculation is worth doing. Résumés went to employers in batches, so you'll need to think about what the unit of assignment really is. And the problem Kohler-Hausmann raises is a real one. No additional data settles it, and that makes it a different kind of difficulty from the rest of this list.

3. What does hormone therapy do for women's health?

Papers.

- Writing Group for the Women's Health Initiative Investigators. Risks and Benefits of Estrogen Plus Progestin in Healthy Postmenopausal Women: Principal Results From the Women's Health Initiative Randomized Controlled Trial. *JAMA* 288(3), 2002.

- The Study of Women's Health Across the Nation (SWAN). Study website. Public-use data at ICPSR: baseline, visit 1, visit 2 and visit 3.

Summary. For a long time, observational research suggested that hormone therapy protected women's health. Then a large randomized trial, the Women's Health Initiative, reported harms, and medical practice changed abruptly. One explanation for the disagreement is that the women who chose hormone therapy were different from the women who didn't. We'll give you data from SWAN, which followed 3,302 women through menopause. None of them were using hormone therapy when they enrolled, and 527 started later, so every woman has symptoms measured before any treatment. SWAN can't reproduce the trial's results, because the trial's outcomes were heart disease, stroke and breast cancer, and the SWAN data we give you measures symptoms, blood pressure and the like. What it does let you do is look directly at how the women who start hormone therapy differ from the women who don't.

Challenges. It's easy to compare the women who started hormone therapy with the women who never did. But that comparison isn't a treatment effect, because women who start aren't like women who don't. You can see how far apart the two groups are by comparing them on everything measured *before* treatment. Then the question is what survives when you compare like with like. Two variables you'd want, education and the clinical site, aren't in the public release. So part of the work is saying precisely what you can't rule out.

4. Can small incentives improve public health?

Papers.

- Abhijit Vinayak Banerjee, Esther Duflo, Rachel Glennerster and Dhruva Kothari. Improving Immunisation Coverage in Rural India: Clustered Randomised Controlled Evaluation of Immunisation Campaigns With and Without Incentives. *BMJ* 340, c2220, 2010. Open access. Data: Harvard Dataverse.

Summary. Immunization rates in rural Rajasthan were very low. To see what would raise them, the researchers randomly assigned 134 villages to one of three groups. Some villages got reliable monthly immunization camps, some got the camps plus a small food incentive, and some got nothing. The data is fully public. There are 74 control villages, 30 with camps and 30 with incentives, and there are records for 5,565 children, each linked to its village.

Challenges. Full immunization is a yes-or-no outcome, so the headline result is a difference in proportions. But villages were assigned, not children, and children in the same village aren't independent. There are thousands of children but only 134 independent assignments. That gap is the center of this project, both for the uncertainty in your estimates and for how large a study needs to be. You'll also have to decide what "fully immunized" means. The authors' code has several definitions and age ranges, and the answer changes depending on which one you use.

5. Are healthcare algorithms measuring the right thing?

Papers.

- Ziad Obermeyer, Brian Powers, Christine Vogeli and Sendhil Mullainathan. Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations. *Science* 366(6464), 2019. Data and code: GitLab.

Summary. A widely used algorithm decided who got extra medical care. It did this by predicting each patient's future medical costs, using cost as a stand-in for how sick they were. If two equally sick people generate different spending, the algorithm ranks them differently, even though it's supposed to be ranking them by need. The authors found that, at the same risk score, Black patients were sicker than white patients, because less was spent on their care. The public data covers 48,784 patient-years. For each one, it has the algorithm's risk score, spending, a count of active chronic illnesses, demographics and lab values.

Challenges. Start with patients the algorithm gave the same score, and ask whether they're equally sick. Comparing averages within groups is enough to see the whole phenomenon. Then test that result. How much does it depend on how you measure "sick"? The data is synthetic, built to imitate confidential records, so you'll have to think about what that lets you claim.

6. Do student evaluations measure good teaching?

Papers.

- Scott E. Carrell and James E. West. Does Professor Quality Matter? Evidence from Random Assignment of Students to Professors. *Journal of Political Economy* 118(3), 2010. Free authors' draft. Code and log: the supplementary material on the article's journal page.
- Michela Braga, Marco Paccagnella and Michele Pellizzari. Evaluating Students' Evaluations of Professors. *Economics of Education Review* 41, 2014. Free working-paper version.

Summary. At the US Air Force Academy, students are randomly assigned to instructors, and then they all take the same follow-on courses. So you can ask whether the instructors whose students get the best immediate grades and give the best evaluations also produce students who do better later. Carrell and West report that they don't. Their data belongs to the Air Force Academy and isn't available to readers. That changes the statistical question rather than removing it. Without the observations, you can't check their estimate, so you have to reason about their *procedure* from the summaries they published. A great deal of research is in this position, and knowing what you can still check is a skill worth having.

Challenges. You get their published tables, their complete analysis code and the log of everything their program printed, including summary statistics for every variable they used.

That's enough to specify a population, with the same number of instructors, the same spread of class sizes and the same variation in student ability. If you simulate that population and run their procedure on it, you know the truth, so you can see how their estimator behaves under a design that matches theirs. You can see its bias and its spread, and whether a study that size could reliably detect an effect as small as the one they report. The study has 34,786 student-course observations but only 421 instructors, and instructors taught anywhere from 1 to 43 sections. How much independent information does a design like that really carry? There's a limit to what this tells you. You can find out how reliable the procedure is at that scale, but you can't check whether they carried it out correctly.

A thread across the six. The same few problems come up across these topics. Some studies withhold their data. Carrell and West released code but no data, Obermeyer's team released synthetic records in place of confidential ones, and the Women's Health Initiative keeps its records behind an application. Sometimes a variable you need isn't in the data you have, like education and clinical site, which aren't in SWAN's public release. And sometimes what was measured isn't the thing you care about, as when spending stands in for illness. A better measure would fix that.

Topic 2 raises a different kind of problem. Kohler-Hausmann's claim isn't that a name is a poor measure of race. It's that "the effect of race" may not be a meaningful quantity at all, and if she's right, no better measure would fix it. Groups on different topics will run into these problems from different sides, so they should compare notes.